

ANAMNESIS: Benchmarking Patient-Level Retrieval-Augmented Generation over Clinical Records on Smartphones

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Abstract

Personal health assistants promise to answer the questions patients ask about their own clinical history, yet existing benchmarks target a single document, adopt multiple-choice formats that bypass retrieval, or omit the evidence annotations needed to diagnose where a retrieval-augmented generation (RAG) pipeline fails. We introduce ANAMNESIS, the first benchmark for patient-level RAG over personal clinical records in document form. Built from real, de-identified MIMIC-III clinical text, ANAMNESIS represents each patient's history as a folder of per-admission PDFs and pairs it with open-ended questions, reference answers, and gold evidence chunks, enabling retrieval and generation to be evaluated separately. The benchmark couples *single-document* questions, reflecting common lookups about a specific clinical episode, with *multi-document* questions that require combining evidence across admissions, including longitudinal questions tracking clinical evolution over time. Because the sensitivity of clinical records makes local execution a requirement rather than an optimization, we accompany the benchmark with a reference system running entirely on a consumer smartphone, where retriever and generator share a single memory budget. We ask how much of the task survives these constraints, comparing lexical, neural, and hybrid retrievers and quantized generators of multiple families and scales.¹

CCS Concepts

• Information systems → Information retrieval; • Applied computing → Consumer health; • Computing methodologies → Natural language processing.

Keywords

Retrieval-Augmented Generation, Question Answering, Benchmark, Electronic Health Records, Longitudinal Clinical Records, On-Device Large Language Models, Edge Computing

ACM Reference Format:

Anonymous Author(s). 2026. ANAMNESIS: Benchmarking Patient-Level Retrieval-Augmented Generation over Clinical Records on Smartphones. In *Proceedings of The 3rd International Workshop on Knowledge-Enhanced Information Retrieval*, co-located with the 35th ACM International Conference on

¹Code and data: <https://anonymous.4open.science/r/anamnesis-keir> (MIT license). The license covers our benchmarking code and the scripts that rebuild ANAMNESIS exactly from credentialed MIMIC-III data; the underlying clinical text is governed by the PhysioNet Data Use Agreement and is not redistributed.

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KEIR '26, Rome, Italy

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Information and Knowledge Management (KEIR '26). ACM, New York, NY, USA, 11 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Patients accumulate clinical documents throughout their lives: discharge summaries, radiology reports, laboratory results, and consult notes. When a question arises—“*When was my last echocardiogram?*”, “*How did my therapy change between my two hospitalizations?*”—the answer is scattered across this growing personal archive, written in a technical register, and rarely at hand when needed. Large language models (LLMs) paired with retrieval-augmented generation (RAG) [6, 12] offer a natural interface to this archive: retrieve the relevant fragments of the patient's own record, then generate a grounded, conversational answer. We refer to this setting as *patient-level RAG*: the retrieval corpus is not an external knowledge base shared across users, but the private, longitudinal document collection of a single individual.

Two properties make this setting distinctive. First, patient questions vary in the scope of evidence they require. Some concern a specific episode of care and can be answered from a single document, whereas others require the connection of evidence across multiple encounters. We refer to these as *single-document* and *multi-document* questions. The latter range from comparing or aggregating facts across encounters to the properly longitudinal case of tracking how a condition, diagnosis, or treatment evolved over time. Second, patient-level RAG is inherently *privacy-critical*: the corpus contains protected health information, making local, on-device deployment a compelling operating point because patients' documents and queries must never leave their smartphones [15]. However, operating entirely on-device imposes strict resource constraints on every component of the pipeline, from model size and embedding computation to vector storage.

Despite rapid progress on clinical question answering (QA), no existing resource evaluates this setting (see §2). Span-extraction datasets such as emrQA [16] and physician-authored question collections such as DiSCQ [11] operate on a single document. Multi-document benchmarks either adopt multiple-choice formats that sidestep retrieval and free-form generation entirely (LongHealth [1], EHRNote-ChatQA [9]), provide a single question per patient without evidence annotations (EHRNoteQA [10]), or are built on access-restricted institutional records (MedAlign [5]). ArchEHR-QA [18] is the only resource with sentence-level evidence ground truth for open-ended answer generation, but it targets excerpts from a single clinical episode. In short, the community lacks a benchmark that *jointly* offers (i) open-ended questions and reference answers, (ii) both single-document and multi-document questions over patient-specific collections, (iii) gold evidence annotations that make retrieval separately measurable, and (iv) clinical documents delivered in the rendered form patients actually receive (PDF files) rather than as raw database text.

Dataset	Source	Access	Document Corpus	Open-Ended Answers	Multi-Doc. per Patient	Longitudinal Questions	Evidence Ground Truth	Multiple Qs per Patient
emrQA [16]	i2b2	Credentialed	✗	✓ ^a	✗	✗	✓	✓
DiSCQ [11]	MIMIC-III	Credentialed	✗	✗ ^b	✗	✗	✓ ^b	✓
MedAlign [5]	Institutional EHR	Restricted (DUA)	✗	✓	✓	✓	✗	✓
LongHealth [1]	Fictional (physician-authored)	Public	✗	✗	✓	✓	✗	✓
EHRNoteQA [10]	MIMIC-IV	Credentialed	✗	✓	✓	✓ ^c	✗	✗
ArchEHR-QA [18]	MIMIC-IV	Credentialed	✗	✓	✗	✗	✓	✗
EHRNote-ChatQA [9]	MIMIC-IV	Credentialed	✗	✗	✓	✓ ^c	✓ ^d	✓
ANAMNESIS (ours)	MIMIC-III	Credentialed	✓	✓	✓	✓	✓	✓

^a Extractive spans projected from expert templates, not free-form abstractive references. ^b Physician-authored questions with textual triggers, but no reference answers. ^c Instances may draw on several discharge summaries, but no question type explicitly targets temporal evolution across encounters. ^d Evidence grounding is probed through auxiliary multiple-choice questions rather than span- or chunk-level annotations.

Table 1: Comparison of ANAMNESIS with existing clinical QA resources for RAG evaluation. *Document Corpus*: records are delivered as rendered documents (one PDF per clinical episode) rather than raw text. *Longitudinal Questions*: answering requires combining information from two or more documents of the same patient. ANAMNESIS is the only benchmark jointly providing a document-form longitudinal corpus, open-ended answers, multi-document questions, and retrieval-diagnostic evidence annotations. Marks denote partial support (see notes).

To fill this gap, we present ANAMNESIS²—a benchmark for patient-level RAG over clinical records. We build ANAMNESIS from MIMIC-III [8], grounding every question in real, de-identified clinical language rather than synthetic text. We draw a population-stratified cohort, materialize each hospital stay (*admission*) as a distinct PDF document, and construct open-ended QA items with supporting excerpts that serve as gold evidence for retrieval evaluation. Both single- and multi-document questions come in near-equal shares from two construction techniques: *rule-based*, where a fixed set of question templates is instantiated for each admission and answered from its structured fields and standard sections, and *LLM-generated*, where a language model writes questions about the free-text content of the document. The single-document family measures the precision floor on the lookups patients ask most often; the multi-document family measures whether a system can locate evidence scattered over several documents of the same patient and combine it into one answer—a capability that no existing resource evaluates.

Alongside the benchmark, we release a reference on-device testbed built on the ExecuTorch LlamaDemo app,³ extended with a full RAG layer. On this testbed, we conduct the first study of patient-level RAG under mobile constraints spanning retrieval strategy and generator family and scale—evaluating retrieval quality, end-to-end answer correctness with calibrated embedding-similarity and LLM-as-a-judge [27] metrics, and on-device efficiency.

Our contributions are as follows:

- **Benchmark.** ANAMNESIS, the first patient-level RAG benchmark over clinical records in document form, with open-ended questions, reference answers, and chunk-level evidence ground truth—including multi-document questions absent from prior resources (§3).
- **Reproducible pipeline.** A fully scripted construction procedure over credentialed MIMIC-III data (deterministic patient selection, PDF materialization, excerpt-conditioned

QA generation), enabling regeneration and extension by any credentialed researcher (§3.3).

- **On-device reference system and study.** An end-to-end smartphone implementation and the first evaluation of patient-level RAG under on-device constraints—spanning retrieval strategies, generator families and scales (§5).

2 Related Work

Question answering over clinical records. Early resources frame clinical QA as machine reading comprehension over a single document. RadQA [19] and emrQA [16] generate questions automatically from annotated clinical and radiology notes; both are large-scale but their questions follow fixed templates and their answers are extracted spans. DiSCQ [11] collects 2,000+ questions that physicians spontaneously asked while reading MIMIC-III discharge summaries, capturing genuine information needs but providing no reference answers. A second wave targets whole patient records. MedAlign [5] asks models to follow clinician-written instructions over longitudinal patient histories (e.g., summarize the last three visits), but records are structured event streams rather than free-text documents, and no evidence annotations are given. LongHealth [1] uses fictional multi-document patient cases designed to stress cross-document retrieval, including distractor documents from other patients, but evaluates through multiple choice. EHRNoteQA [10] generates one question per patient over multiple discharge summaries and supports both open-ended and multiple-choice evaluation, but provides no evidence ground truth. EHRNote-ChatQA [9] scales to multi-turn dialogues and pairs each content question with an evidence-grounding question identifying its supporting source, but answers are multiple choice throughout. Closest to our protocol, ArchEHR-QA [18] supports open-ended answers scored by sentence-level citations, yet each case covers a single clinical episode. As summarized in Table 1, no prior resource combines a document-form corpus, open-ended answers, multi-document questions, and evidence annotations. ANAMNESIS provides all four.

²From the Greek *anámnesis* (“recollection”): in medicine, the patient’s clinical history as reconstructed from their account and records—precisely what our benchmark asks systems to navigate.

³<https://github.com/meta-pytorch/executorch-examples>

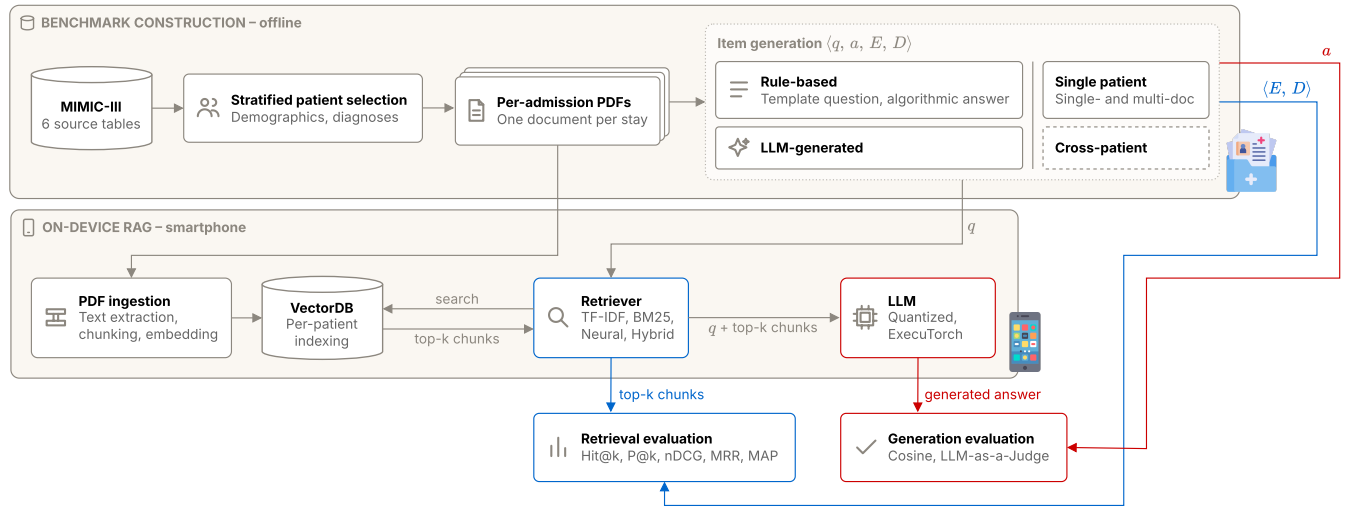


Figure 1: ANAMNESIS at a glance. Credentialed MIMIC-III admissions are materialized as chronological patient folders, while QA instances pair field, narrative, or longitudinal questions with gold evidence and reference answers. A quantized on-device LLM is evaluated under no-context, full-folder, retrieved top-3 (RAG), and gold-evidence (oracle) conditions. Separate retrieval and answer evaluation enables error attribution between retrieval and generation.

Synthetic and generated clinical benchmarks. Because real records cannot be freely shared, LLM-assisted generation over de-identified or synthetic substrates has become a standard recipe for clinical benchmarks, including LongHealth [1] (fictional multi-document patient cases for long-context QA), EHRNoteQA and EHRNote-ChatQA [9, 10] (QA over real discharge summaries), and Paint It, Black [25] (synthetic clinical PDFs for de-identification evaluation). ANAMNESIS follows this line but anchors questions in *real* de-identified clinical text: the gold evidence is verbatim text from actual MIMIC-III notes, keeping the benchmark grounded in authentic clinical language rather than fictional surrogates.

On-device LLMs and RAG. Quantization and dedicated runtimes now enable local execution of billion-parameter LLMs on consumer smartphones, and a growing body of work characterizes this regime [7, 13, 22, 24]. In the clinical space, early systems explore on-device LLMs for decision support [15] and privacy-preserving mental-health assistance [3, 23]. Yet, these efforts measure speed, memory, or generic quality; none evaluates retrieval-grounded generation over a private clinical document collection. To our knowledge, ANAMNESIS is the first resource enabling that evaluation.

3 The ANAMNESIS Benchmark

This section details the construction and composition of ANAMNESIS (Fig. 1). Our design follows four principles: **(1) Patient-centric organization.** The unit of the benchmark is the patient, not the individual document: each instance is evaluated against the patient’s entire document folder, so retrieval is always performed over a realistic personal archive. **(2) Retrieval-diagnostic supervision.** Every question carries the identity of the evidence chunk(s) that generated it, decoupling retrieval failures from generation failures by construction. **(3) Multi-document pressure.** A dedicated question family requires combining evidence across multiple documents of the same patient, making cross-document retrieval necessary

rather than incidental, while single-document questions keep the benchmark representative of the everyday lookups patients actually pose. **(4) Reproducibility.** All selection and generation steps are deterministic given released identifiers and seeds, so any credentialed MIMIC-III user can regenerate or extend the benchmark.

3.1 Source Data and Patient Selection

ANAMNESIS builds on MIMIC-III [8], a de-identified critical-care database whose NOTEVENTS table stores free-text clinical notes linked to patients and admissions across 15 documentation types. We prefer MIMIC-III over MIMIC-IV because it underpins the QA resources we compare against (emrQA, DiSCQ), easing cross-resource analysis, and because its single note table will allow future versions of ANAMNESIS to extend beyond the current document type without changing corpora. The present release uses *discharge summaries only*: they are produced for every stay and routinely given to the patient; they integrate the whole admission into a self-contained narrative suitable for both single-document lookups and cross-admission comparisons; and they carry the structured fields that rule-based items can be verified against. The full rationale and MIMIC-III table schema are provided in the linked repository. When an admission has several discharge-summary notes, they are concatenated in source-table order so each admission yields exactly one document.

We select 100 patients via nested stratified sampling. The eligible population is first split by the presence of a mental-health (MH) diagnosis (ICD-9 chapter 290–319), and patients are drawn from *both* groups in proportion to their base rate in MIMIC-III—ensuring the MH subgroup is represented at its true prevalence rather than by chance, without restricting the benchmark to that population. We stratify on this axis for two reasons. First, MH patients are disproportionately likely to rely on conversational AI for health information: around a quarter of surveyed U.S. adults

already use chatbots for MH support [14], and stigma around seeking in-person treatment makes privacy—and therefore on-device execution—a primary driver of that preference [4]. Second, a personal health assistant that systematically under-represents this subgroup would fail precisely the users who depend on it most. The inner stratum balances sex, age quintile, and note-length quintile. All admissions of every selected patient are retained, yielding 138 admissions (1–6 per patient); patients with a single admission contribute single-document questions only, while 22 patients with two or more admissions anchor the multi-document set (§3.4). Identifiers are released, making the cohort exactly reconstructable.

3.2 Document Materialization

For each admission, we render the associated discharge-summary text into a single PDF document; the documents of the same patient are then collected into one folder. The PDF opens with a fixed structured header that describes the admission, followed by a free-text body carrying the discharge-summary narrative itself (e.g., history of present illness, hospital course, and discharge instructions). Header fields include patient demographics, admission type, the clinical service responsible for the admission (e.g., cardiology, general surgery), admission and discharge dates, and principal diagnosis; every field is either copied verbatim from the MIMIC-III tables and columns or derived deterministically from them. Documents are rendered with ReportLab⁴ on A4 pages. A condensed two-page example is shown in the linked repository.

3.3 Question–Answer Generation

Each evaluation item is a tuple $\langle q, a, E, D \rangle$: a question q , a reference answer a , the gold evidence chunk(s) E (verbatim excerpts from the corpus), and D , the set of admissions whose evidence is required.

Single-document items. A single-document item has $|D| = 1$; hence, all gold evidence lies within one PDF. The set is split evenly between the two construction techniques. **(1) Rule-based:** each item is produced by filling a *template*—a hand-written question skeleton with a single variable slot identifying the target admission. The slot can be filled with the exact start date, a coarser reference, or a relative reference, mirroring the imprecise way patients naturally refer to past hospitalizations (e.g., “What medications was I prescribed when I was discharged from my hospital stay in [month/year]?”). Answers are derived *algorithmically*, never by an LLM. We organize templates into six types: (i) *header facts*—direct fields such as principal diagnosis, admission type, or admission date; (ii) *derived header facts*—information that can be deduced from the header, including age and length of stay; (iii) *section lookups*—passages from standard discharge-summary sections; (iv) *presenting complaint*—what symptoms brought the patient to hospital; (v) *allergy status*—Yes/No questions over the Allergies section, relevant whenever a patient must confirm their allergy history to a new clinician; (vi) *admission disambiguation*—type-(i) questions, but posed for a patient with multiple hospitalizations and naming the other admission dates explicitly, making retrieval more challenging. Items are drawn via a global max-flow assignment that enforces an equal number of instances per template type. **(2) LLM-generated:** items

are authored by Claude Sonnet 5, accessed through the Anthropic Messages API. Each request is stateless and self-contained: it carries only the instruction and the full text of one document, which fits comfortably within the model’s 1M-token context window, so no document is truncated and no history or memory is shared across admissions. The prompt template asks for exactly one item—an open-ended question about the free-text narrative (clinical events, findings, procedures, medications), its reference answer, and the exact supporting passage as `source_chunk`—returned as a JSON object; two independent requests are issued per admission. The passage is model-selected but constrained to be a verbatim substring of the input, which is verified downstream (see “Quality control” below). Decoding follows the provider’s fixed defaults.

Multi-document items. A multi-document item has $|D| \geq 2$, with all admissions in D belonging to the same patient: the gold evidence is spread over several PDFs and the answer requires combining it. Items subdivide by $|D|$ and, within each sub-type, are again produced using both rule-based and LLM-generated techniques. **Two-admission** ($|D| = 2$): evidence spans exactly two admissions of a patient. The rule-based technique instantiates, for every admission pair, one comparison template over a structured header field asking for the value at each admission and whether it stayed the same or changed; the field is chosen greedily so that, across the whole set, the fields and their two outcomes remain as evenly represented as possible, and each admission’s header chunk is annotated as gold evidence ($|E| = |D|$). The LLM-generated technique first identifies a clinical entity—a diagnosis, medication, or procedure mention—that recurs in both documents, and samples from each document the excerpt surrounding it. The two excerpts and the shared entity are then submitted to Claude Sonnet 5, under the same configuration as above, with the prompt requiring a question that connects the entity across the two admissions and cannot be answered from either excerpt alone. The model returns, for each excerpt it relies on, the verbatim substring supporting the answer; these substrings form E and are verified as for single-document items. **Trajectory** ($|D| \geq 3$): evidence spans three or more admissions of the same patient in chronological order, and the question asks how a fact evolved across them. The rule-based technique applies the same comparison templates to consecutive admission triples only, so that the temporal reading stays unambiguous; the LLM-generated technique receives one excerpt per admission and must produce a question whose answer traces the evolution across all of them. Because these questions are inherently about change over time, they form the properly *longitudinal* subset of ANAMNESIS.

Cross-patient probe (auxiliary). ANAMNESIS assumes one folder per patient, and the reference system enforces this partition by construction. In practice the assumption can be violated: a single device may hold the records of several people, as when a caregiver manages a relative’s documents alongside their own. To measure how a system behaves in this setting, both construction techniques also produce items that span two *different* patients, always verified to draw evidence from both. A pair of admissions is selected so that the two patients share the same principal diagnosis; the rule-based technique then asks for the same *unrelated* header field for each and whether it coincides (e.g., “Both patients were admitted for congestive heart failure: which hospital service treated each, and was it the

⁴<https://www.reportlab.com/opensource/>

Question Family	RB	LLM	Total
Single-document	276	276	552
Multi-document	81	68	149
Two-admission ($ D = 2$)	65	62	127
Trajectory ($ D \geq 3$)	16	6	22
Patient-level benchmark	357	344	701
Cross-patient probe (<i>auxiliary</i>)	35	32	67
Corpus	Value		
Patients (with ≥ 2 admissions)	100 (22)		
Admissions = documents	138		
per patient (min / avg / max)	1 / 1.38 / 6		
words per document (min / avg / max)	457 / 1,619 / 5,966		

Table 2: ANAMNESIS composition, corpus, and per-set item statistics. Multi-document items involve a single patient, with sub-types defined by the number of admissions $|D|$ whose evidence is required. The cross-patient probe (below the dashed rule) is released alongside the benchmark but excluded from patient-level benchmark scores. Multi-document items are constructed from patients with at least two admissions. RB: rule-based; LLM: LLM-generated.

Item statistics	Single-patient				Cross-patient	
	Single-doc.		Multi-doc.		RB	LLM
Items per patient (min/avg/max)	2/2.8/12	2/2.8/12	1/3.7/19	1/3.2/15	1/2.5/6	1/2.3/4
Documents per item (min/avg/max)	1/1/1	1/1/1	2/2.2/3	2/2.1/4	2/2/2	2/2/2
Gold chunks per item (avg)	1.0	1.0	2.2	2.2	2.0	2.0
Question words (min/avg/max)	12/18.3/27	6/15.4/27	22/23.6/29	9/25.5/39	31/32.7/37	13/23.7/40
Answer words (min/avg/max)	1/7.3/76	1/20.7/78	17/21.9/31	18/33.7/53	21/23.8/28	18/34.3/47

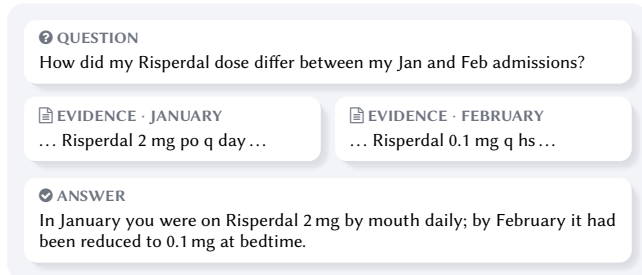


Figure 2: A real ANAMNESIS multi-document item (two-admission, LLM-generated, $|D| = 2$). Each gold evidence chunk is a verbatim excerpt of one admission; the answer cannot be assembled from either document alone.

same?”), evenly balanced across five such fields. The LLM-generated technique instead asks for a narrative detail of the *same* shared finding (e.g., “Both patients were hospitalized for congestive heart failure. How did their ejection fractions compare?”), using the same multi-document prompt template; this subset naturally reflects the most prevalent shared diagnoses supported by the underlying cohort, rather than an artificially balanced topical distribution. The shared diagnosis makes the two records comparable, and the answer must be assembled from one document per patient, so for these items only the index is built over the union of the two folders. What the probe measures is therefore two conditions of a shared archive: whether retrieval reaches both folders, and whether the answer attributes each fact to the right person. Because these questions do not correspond to any patient’s own information need, they are excluded from the benchmark and its scores and released as an auxiliary set, evaluated with the contamination diagnostic of §4 applied at the patient rather than the admission level.

Quality control. Items pass three automatic checks. (i) **Answerability:** rule-based templates fire only when the target field or section is populated; any item whose evidence cannot be located on a verified page is discarded. (ii) **Exact-substring grounding:** every gold chunk must appear verbatim in the text re-extracted from the corresponding PDF, guarding against LLM paraphrasing and

against mismatches between the PDF extractor used at generation time (pypdf) and the one used on device (PDFBox). (iii) **Answer-evidence overlap:** a lexical heuristic flags items whose reference answer shares fewer than four content words (case-folded, >3 characters) with its evidence chunk for manual review; flagged items are overwhelmingly short numeric or list-style answers (e.g., a lone lab value or date) that legitimately share little vocabulary with their source sentence, not hallucinations, and are kept after review. To complement the automatic checks, all 344 LLM-generated benchmark items were reviewed by the authors. For multi-document items, a *single-excerpt insufficiency* check is enforced at generation time: the prompt requires the model to verify, per excerpt, that no single excerpt alone would let someone answer the question completely, and to decline (return null) rather than force an item when no genuine multi-document question can be composed.

3.4 Benchmark Statistics

Table 2 gives the composition and distributional statistics of ANAMNESIS. The primary patient-level benchmark contains 701 items, split roughly evenly between the two construction techniques (357 rule-based, 344 LLM-generated), so no comparison across families is confounded by generation method. The 22 patients with two or more admissions all contribute multi-document items, spanning 60 admissions and on average 2.2 documents per item. Multi-document items are longer and more demanding than single-document ones both in question complexity (24.5 vs 16.9 words on average) and expected answer length (27.3 vs 14.0 words). The 67-item cross-patient probe is released alongside the benchmark but excluded from all primary evaluations. Fig. 2 shows a multi-document item.

4 Experimental Setup

ANAMNESIS is deployment-agnostic: any RAG pipeline can be evaluated against it. To instantiate the mobile deployment setting that motivates the benchmark, we provide a reference implementation of the full RAG pipeline as an Android application and evaluate it on a consumer smartphone. We first describe the execution environment and local document store, followed by the retrieval and generative models, evaluation protocols, and implementation details.

4.1 Target Execution Environment

The reference implementation is a Kotlin Android application that performs document ingestion, indexing, retrieval, and generation without a server component. The patient’s documents and questions therefore remain on the device. The implementation is available in the linked repository. All on-device experiments are conducted on a Realme RMX3081 with approximately 5.6 GB of usable RAM.

The memory constraint applies to the complete pipeline rather than to either model in isolation: the generator, embedding model, vector index, retrieved text, and runtime buffers must coexist within the device’s available memory. We thus report the on-disk sizes of the generator and embedding model separately, but measure peak resident set size (RSS) for the complete pipeline configuration. The benchmark does not impose a fixed per-model size limit. A configuration is considered on-device deployable if all its components can be loaded concurrently and it can complete end-to-end inference on the reference device without exceeding the available memory.

4.2 Local Document Store and Ingestion

Uploaded PDFs are parsed locally using PDFBox-Android v2.0.27.0. All PDFs used in the benchmark contain an embedded text layer, so direct text extraction is sufficient and the OCR fallback is not exercised for benchmark ingestion. This design intentionally isolates the RAG pipeline from additional document-processing challenges such as OCR errors or complex scanned layouts, allowing the evaluation to focus on ingestion, chunking, retrieval, and generation over document-based clinical records. Extracted text is processed in page order and normalized by collapsing consecutive spaces and tabs into a single space and trimming leading and trailing whitespace, while preserving line breaks.

Chunking is character-based. Starting from the current position in the text, the splitter considers at most C characters and selects the last available paragraph boundary within that span. If no paragraph boundary is available, it selects the last line boundary; if neither exists, it performs a hard split at C characters. Consecutive chunks overlap by O characters. Thus, C is an upper bound rather than a fixed chunk length: a chunk may be shorter when an earlier paragraph or line boundary is selected. We compare $C \in \{500, 1000, 1500\}$ and select $C = 500$ with $O = 100$ for the remaining experiments (see §5.1).

Chunks, embeddings, and their source metadata are persisted in ObjectBox 4.0.1,⁵ an embedded database providing local HNSW-based approximate nearest-neighbor search. ObjectBox is the only vector store used in the experiments. It was selected because it supports persistent storage of vectors and associated metadata within an Android application without requiring an external database service. Each chunk is stored together with its source-document identifier, admission identifier, and position within the document. These fields allow retrieved chunks to be mapped back to their source PDFs; document information is stored as metadata rather than prepended to the chunk text. The application returns this metadata with each retrieved chunk and displays it to the user.

⁵<https://docs.objectbox.io/on-device-vector-search>

4.3 Retrieval Models

All retrievers search the complete folder of the patient associated with each question. Consequently, every chunk that does not belong to the annotated evidence constitutes a distractor, including chunks from other admissions of the same patient.

We evaluate two lexical retrievers, TF-IDF and BM25 [17], and three dense retrievers, EmbeddingGemma [20], multilingual-e5-small [21] and granite-embedding-311m-multilingual-r2 [2]. For encoders that distinguish between query and document inputs (i.e., EmbeddingGemma), we apply the model-specific prefixes prescribed by their authors.

We additionally evaluate a hybrid configuration combining BM25 and EmbeddingGemma through reciprocal rank fusion:

$$s_{\text{RRF}}(x) = \frac{1}{c + \text{rank}_{\text{BM25}}(x)} + \frac{1}{c + \text{rank}_{\text{EmbeddingGemma}}(x)} \quad (1)$$

where x denotes a candidate chunk and $c = 60$.

For lexical and dense retrieval, chunks are ordered by decreasing retrieval score; for hybrid retrieval, they are ordered by decreasing RRF score. The first three chunks are concatenated in that order and supplied to the generator together with the question.

4.4 Generative Models

We evaluate multiple instruction-tuned generators, spanning different model families and parameter scales: Qwen3-0.6B/1.7B/4B, Llama-3.2-1B, and Phi-4-mini.

All on-device generators are exported to the .pte format via the same ExecuTorch pipeline, using the XNNPACK backend and 8da4w quantization, with 4-bit weights and dynamic 8-bit activations.

For generator comparison, BM25 is held fixed as retriever. Each generator is assessed under four evidence conditions: **Closed-book**, without any record context; **Full-context**, with the complete patient folder when it fits within the model’s context window; **RAG**, with the top three chunks returned by BM25; and **Oracle evidence**, with the annotated gold chunks. The closed-book condition measures whether the model can answer without access to the patient’s records, whereas the oracle condition estimates its generation ceiling when the required evidence is provided directly. The difference between the RAG and oracle conditions captures the effect of retrieval failures and distractor evidence on answer quality.

Generation uses a neutral factual-QA prompt that instructs the model to answer concisely using only the supplied clinical notes and to state explicitly when the requested information is absent. The prompt is formatted using the model’s ChatML template. Decoding is greedy, with temperature 0 and a maximum of 768 new tokens.

4.5 Retrieval Evaluation

Let G_q denote the set of gold evidence chunks associated with question q . Because evidence annotations and runtime chunks may have different boundaries, relevance is determined through token overlap. A retrieved chunk r covers a gold chunk $g \in G_q$ when:

$$\text{overlap}(r, g) = \frac{|T_r \cap T_g|}{\min(|T_r|, |T_g|)} \geq 0.5 \quad (2)$$

where T_r and T_g are the sets of case-folded alphanumeric tokens in r and g , respectively. This containment-based criterion tolerates

	Retriever	Hit@1	Hit@3	Hit@5	P@3	P@5	P@10	nDCG@10	MRR	MAP
Single Doc	TF-IDF	35.7	50.0	59.1	21.7	16.7	11.4	52.0	46.3	43.1
	BM25	50.9	70.7	81.3	31.2	23.5	14.7	67.9	63.1	58.7
	EmbeddingGemma	<u>54.9</u>	<u>75.9</u>	<u>85.0</u>	<u>34.2</u>	<u>26.2</u>	<u>16.1</u>	<u>72.2</u>	<u>67.5</u>	<u>63.6</u>
	multilingual-e5-small	42.2	64.7	75.2	27.7	20.8	13.6	62.2	55.9	52.9
	Granite-311M-R2	50.0	72.1	81.2	33.2	25.4	<u>16.1</u>	68.8	63.2	59.6
	Hybrid (BM25 + EmbeddingGemma)	63.0	80.8	87.3	39.6	28.3	16.6	73.8	73.7	67.2
Multi-Doc	TF-IDF	32.4	54.4	68.5	26.8	23.4	18.3	54.9	47.1	42.8
	BM25	<u>64.9</u>	<u>79.6</u>	<u>84.9</u>	54.1	43.5	29.5	<u>75.8</u>	<u>73.5</u>	<u>68.2</u>
	EmbeddingGemma	47.1	73.4	82.4	42.3	36.4	26.5	67.9	61.7	57.6
	multilingual-e5-small	30.1	51.7	63.1	22.5	19.1	15.5	49.6	43.4	38.8
	Granite-311M-R2	63.0	77.4	81.8	<u>50.7</u>	<u>41.8</u>	<u>28.1</u>	72.8	71.2	64.4
	Hybrid (BM25 + EmbeddingGemma)	71.7	87.0	93.1	44.3	31.7	18.2	78.7	80.7	71.9

Table 3: Retrieval performance on ANAMNESIS. Best scores in bold, second-best underlined.

boundary differences while requiring at least half of the shorter chunk to be shared.

Let $R_k(q)$ be the top- k retrieved chunks. We define the gold evidence covered within the first k results as follows:

$$C_k(q) = \{g \in G_q : \exists r \in R_k(q) \text{ such that } \text{overlap}(r, g) \geq 0.5\} \quad (3)$$

We compute all cutoff-based metrics at the same depths, $k \in \{1, 3, 5\}$. Hit@ k indicates whether at least one gold chunk is covered, Evidence Coverage@ k is $|C_k(q)|/|G_q|$, and Full-Evidence Hit@ k indicates whether $C_k(q) = G_q$. We additionally report nDCG@ k , using binary relevance for retrieved chunks that cover at least one gold chunk, and MRR, based on the rank of the first such chunk.

The main comparison reports all cutoff-based metrics at $k = 3$, matching the number of chunks supplied to the generator. Results at $k = 1$ measure first-rank retrieval, while $k = 5$ provides a limited-headroom analysis beyond the generation budget. The same definitions are applied to single- and multi-document questions. Because each single-document item contains one gold evidence chunk, Hit@ k , Evidence Coverage@ k , and Full-Evidence Hit@ k coincide in that setting. For multi-document questions, the three metrics distinguish retrieving any evidence, a subset of the required evidence, and all required evidence.

4.6 Answer Evaluation

Generated answers are evaluated against their reference answers using one embedding-based measure and one LLM judge. For the embedding-based evaluation, we compute the cosine similarity between the generated and reference answers in the embedding space of Jasper-Token-Compression-600M [26]. An answer is counted as passing when its cosine similarity is at least 0.75.

For the judge-based evaluation, GPT-5.6 Sol receives the question, reference answer, and generated answer. The judge classifies the generated answer as CORRECT, PARTIALLY CORRECT, or INCORRECT according to a rubric that places particular emphasis on negation and the accuracy of numeric values and dates. The complete evaluation prompt is provided in the repository.

4.7 Implementation Details

Generative-model inference is performed on the CPU using ExecuTorch 1.1.0 with the XNNPACK backend. Dense embedding inference uses Google LiteRT 1.4.2 or MediaPipe Tasks Text 1.0.0, depending on the model’s export format. ObjectBox 4.0.1 provides persistent storage for chunks, embeddings, and source metadata, as well as HNSW-based vector search.

All on-device latency and memory measurements are collected directly through the application’s instrumentation harness on the reference smartphone. Latency is measured as wall-clock time, while memory usage is reported as peak process RSS. The boundaries of each timed operation are specified alongside the results.

5 Results

5.1 Chunking Granularity

Chunk granularity affects both retrieval quality and on-device index size. As shown in Table 4, 500-character chunks achieve the best Hit@1, nDCG@10, MRR, and MAP, while 1000-character chunks yield a slightly higher Hit@5 and fewer chunks per document. Given the stronger performance at the highest ranks, 500-character chunks with 100-character overlap are used in the next experiments.

5.2 Retrieval Comparison

Table 3 first compares all retrieval families over full patient folders. Three observations stand out. (1) **Dense retrieval is the strongest single family on this task.** EmbeddingGemma reaches Hit@1 of 54.9% against BM25’s 50.9% and TF-IDF’s 35.7%, and leads on every other cutoff and ranking metric. This is consistent with clinical questions being phrased in natural language rather than record terminology, which favors semantic over literal token matching, and with many gold chunks being short, templated fields, where BM25’s term-frequency and length-normalization statistics behave less reliably than on longer prose. (2) **Granite and multilingual-e5-small** trail EmbeddingGemma but remain competitive with or ahead of BM25 on most columns, so the advantage is not specific to a single encoder. (3) **Fusing BM25 and EmbeddingGemma** via reciprocal rank fusion (Hybrid) is the best configuration overall,

Chunk Size	Hit@1	Hit@3	Hit@5	nDCG@10	MRR	MAP	Chunks/Doc
500 chars	73.2	87.0	89.9	79.6	80.9	68.2	29.1
1000 chars	68.8	87.0	92.8	76.8	78.9	64.8	12.8
1500 chars	66.7	85.5	89.9	77.0	76.9	67.0	8.4

Table 4: Neural retrieval as a function of chunk size.

Generator	Retriever	LLM-as-a-Judge			Cosine	
		Correct (%)	Partial (%)	Incorrect (%)	Pass	Agr.
Qwen3-1.7B	TF-IDF	34.8	18.3	46.9	58.1	69.2
	BM25	48.1	21.5	30.4	59.0	73.9
	EmbeddingGemma	41.4	24.0	34.6	57.2	66.6
	Hybrid	50.8	21.1	28.1	66.9	65.6

Table 5: End-to-end answer quality by retriever with Qwen3-1.7B held fixed as generator. Judge verdict rates (%), cosine-similarity pass rate at threshold 0.75, and cosine/judge agreement (Agr.). The full generator sweep under all four evaluation conditions is in Table 6.

improving Hit@1 by 8.1 points over EmbeddingGemma alone (63.0% vs. 54.9%) and leading on every other metric, indicating the two retrievers make partially complementary errors.

Multi-document questions. Multi-document questions produce a different retrieval regime from the single-document setting, with substantial changes in both absolute performance and model ranking. BM25 becomes the strongest standalone retriever, leading the other standalone retrievers on every metric, whereas multilingual-e5-small—a mid-table performer on single-document questions—becomes the weakest retriever across the board. The same length-normalization effect plausibly explains the reversal: the excerpts in multi-document items are longer, less templated passages than many single-document header fields, a regime where BM25’s statistics are more reliable, while Granite’s semantic matching keeps it close behind.

5.3 End-to-End Answer Quality

Table 5 evaluates how retrieval quality propagates to answer quality by holding the generator fixed to Qwen3-1.7B and varying the underlying retriever. Because the generator receives only the top three retrieved chunks, the composition of this small evidence set is particularly important. Hybrid retrieval produces the highest rate of correct answers (50.8%) and the highest cosine-similarity pass rate (66.9%), consistent with its strong top-three retrieval performance. BM25 follows with 48.1% correct answers, whereas EmbeddingGemma reaches 41.4% despite outperforming BM25 on several pooled retrieval metrics. Standalone retrieval improvements therefore do not translate directly into better generated answers: beyond retrieving a relevant chunk, successful generation depends on whether the limited top-three context contains sufficiently complete, specific, and non-distracting evidence. Hybrid retrieval is beneficial because lexical and semantic signals recover complementary evidence, increasing the probability that the context supplied to the generator contains the information required for answering.

These results establish that retrieval quality matters end to end, but they do not by themselves determine how much of the remaining error is attributable to retrieval rather than generation.

Generator	Params	Judge Correct (%)			
		Closed-Book	Full-Context	RAG	Oracle Evidence
Qwen3-0.6B	0.6B	9.6	43.5	56.2	62.8
Qwen3-1.7B	1.7B	6.6	42.9	65.3	69.2
Qwen3-4B	4B	4.9	43.6	70.7	75.3
Llama-3.2-1B	1B	4.7	43.2	61.9	68.5
Phi-4-mini	3.8B	5.1	43.0	59.5	64.9

Table 6: Generator sweep under the four evaluation conditions (BM25 retrieval fixed for the RAG column). Judge-Correct rates are obtained using GPT-5.6 Sol (§4). The Oracle-RAG gap isolates the retrieval cost per model; the Oracle column itself bounds the generation ceiling.

5.4 Generator Comparison

Table 6 separates retrieval limitations from generation limitations across five generators and four evidence conditions. For every generator, correctness increases monotonically from Closed-Book to Full-Context to RAG to Oracle Evidence. Closed-book scores are uniformly low, consistent with negligible parametric leakage of record-specific facts, while the large improvement from Full-Context to RAG shows that selecting a small relevant context is more effective than exposing the model to the complete patient folder.

Oracle Evidence improves correctness further for every model, confirming that imperfect retrieval still imposes an end-to-end cost. However, the Oracle-RAG gaps are relatively limited—3.9 percentage points for Qwen3-1.7B and 4.6 points for Qwen3-4B—while even Oracle correctness remains below 76%. Retrieval is therefore not the sole bottleneck: even when the relevant evidence is supplied directly, models still fail to interpret or express it correctly.

Generator capacity accounts for part of this residual difficulty. For Qwen3, both RAG and Oracle correctness increase with model scale, and the Oracle ceiling rises monotonically. The Oracle-RAG gap is smaller for the two larger Qwen3 models than for Qwen3-0.6B, although it does not shrink further from 1.7B to 4B. Full-Context correctness, by contrast, remains nearly flat across all generators despite differences of more than 20 points under RAG and Oracle Evidence. Llama-3.2-1B also trails Qwen3-1.7B under both conditions. Together, these results reveal a coupled retrieval-generation bottleneck: retrieval determines whether useful evidence reaches the model, while generation capacity determines how effectively that evidence is converted into a correct answer.

5.5 Cross-Patient Results

Table 7 reports retrieval performance on the 67-item cross-patient probe (see §3.3). For each item, the retrieval index is constructed over the union of the two patients’ folders.

Although the index spans the union of two patient folders, thereby increasing the total number of candidate documents and potential distractors, the cross-patient items specifically target shared primary diagnoses where semantic or lexical overlap between the two matched records is high. Despite this more challenging retrieval setting, the relative behavior of the retrievers remains broadly consistent with that observed in Table 3: neural retrievers generally

Retriever	Hit@3	P@3	nDCG@10	MRR	MAP
TF-IDF	65.8	19.5	38.7	58.4	32.7
BM25	79.4	<u>44.7</u>	<u>77.0</u>	75.5	<u>70.0</u>
EmbeddingGemma	<u>83.9</u>	46.8	80.1	78.9	72.4
E5-small	86.3	36.2	74.4	70.2	64.1
Granite	<u>83.9</u>	39.4	76.5	<u>76.9</u>	67.1

Table 7: Cross-patient probe retrieval performance ($n = 67$; rule-based and LLM-generated items averaged).

Question Type	n	BM25	EmbeddingGemma
Section Lookup	155	42.6	51.0
Derived/Numeric Fact	98	65.3	65.3
Negation	80	57.5	75.0
Admission Disambiguation	68	42.6	25.0
Header Fact	66	60.6	65.2
Presenting Complaint	46	17.4	17.4
Clinical Course	39	71.8	82.1

Table 8: Single-document Hit@1 by question type (BM25 vs. EmbeddingGemma, full single-document set, $n = 552$).

achieve the strongest overall performance, while BM25 remains a competitive lexical baseline.

5.6 Per-Question-Type Retrieval Breakdown

Table 8 disaggregates single-document Hit@1 by question type, covering the full 552-item single-document set (rule-based + LLM-generated), mapped onto the benchmark’s six rule-based question types (§3.3, 46 rule-based items each) plus *Clinical Course*, an LLM-generated-only category (e.g., *numeric_temporal* is mapped to derived header facts; *section_specific/clinical_detail* to section lookups; *negation_presence* to negation checks). Rule-based generation has no Clinical Course counterpart because its answers must be derived algorithmically from a template and a structured field or fixed section, whereas identifying a trigger event and its causal escalation in free text requires semantic judgment only the LLM-generation track provides.

Admission Disambiguation is the one category where BM25 leads (42.6% vs. 25.0%): these questions name the other admission dates explicitly as distractors (§3.3), and matching an exact date is a literal-token task BM25 is well suited for. Negation shows the largest gap in EmbeddingGemma’s favor (75.0% vs. 57.5%). Presenting Complaint is the hardest category for both retrievers alike (17.4% each)—open-ended questions about how a patient’s clinical story begins, with no fixed anchor phrase to match against. Clinical Course, despite requiring the most synthesis to *answer*, is the easiest category to *retrieve* for both retrievers (71.8% vs. 82.1%): its questions are anchored to a specific, rare trigger detail (an exact fever value, a named drug switch) that is unusually easy to match against, a case where retrieval and answering difficulty diverge. Derived/Numeric Fact and Header Fact are close between the two retrievers, consistent with these being short, structured fields.

Retriever	Rule-Based Hit@1	LLM-Generated Hit@1
BM25	42.8	59.1
EmbeddingGemma	43.1	66.7

Table 9: Single-document Hit@1 by construction technique (BM25 vs. EmbeddingGemma, $n = 276$ each split).

Store	Index (MB)	Indexing (s)	p50 Query (ms)	p95 Query (ms)	Recall@10 vs. Exact
ObjectBox (HNSW)	16.2	3.50	24	50	86.8
SQLite + sqlite-vec	16.0	1.83	14	20	100
USearch (JNI)	12.4	1.98	0.61	0.79	96.1
Brute-force (Kotlin)	13.83	0.64	22	74	100

Table 10: Embedded vector store comparison on device over the full ANAMNESIS corpus (4022 chunks, 768-dim embeddings). Query latency is reported as the median (p50) and 95th percentile (p95). Recall@10 is measured against exact brute-force search.

5.7 Rule-Based vs. LLM-Generated Questions

Table 9 splits Table 3’s pooled single-document Hit@1 by construction technique. The two retrievers are nearly tied on rule-based questions, but EmbeddingGemma pulls ahead by 7.6 points on LLM-generated ones—the dense-retrieval advantage reported in §5.2 is concentrated in the LLM-generated half of the benchmark, consistent with rule-based questions being built from fixed templates over structured fields and LLM-generated ones being phrased more freely in natural language.

5.8 On-Device Storage Efficiency

Table 10 shows the storage side of the on-device constraint. ANN recall loss is not negligible: ObjectBox’s HNSW index reaches only 86.8 Recall@10, below both exact stores (100) and USearch (96.1), while also having the highest query latency (p50 24 ms, p95 50 ms)—not the fastest option on either axis. Brute-force is cheapest to build (0.64 s) and second-smallest on disk, consistent with a 100-patient folder still being small enough for exact search to be competitive.

5.9 Qualitative Error Analysis

Retrieval systems in clinical RAG pipelines exhibit distinct failure modes in which the retrieved context does not contain the evidence needed to answer the question. The following examples illustrate recurring retrieval errors rather than cases in which correctly retrieved evidence is subsequently misinterpreted (see the visual examples in the linked repository):

- **Weak-anchor retrieval failure:** The gold chunk was retrieved within the top 5, but an unrelated hospital-course passage outranked it at rank 1. Short Yes/No questions like this one carry almost no distinguishing content words, leaving matching with little to discriminate on, so an unrelated but token-overlapping passage can outrank the correct field.
- **Semantic-attractor retrieval failure:** While the gold evidence correctly identifies the patient’s specific presentation,

the retriever erroneously ranks an irrelevant electrocardiographic finding as the top result. This generic chunk acts as a semantic attractor, repeatedly surfacing across admissions because its clinical phrasing closely mimics open-ended narrative queries regardless of the actual patient context.

- **Cross-admission contamination:** The retriever found the correct evidence for a different question about this same patient’s earlier admission; the two admissions share enough vocabulary for lexical matching to conflate them.

6 Discussion

Why these retrieval results? As shown in Table 3, on single-document questions, EmbeddingGemma leads but BM25 trails by only 4 points at Hit@1 (54.9% vs. 50.9%); on multi-document questions, BM25 leads but EmbeddingGemma and Granite trail by only a few points on most columns. Neither family is ever decisively ahead, and multi-document questions narrow the gap further, consistent with these items requiring a fact to be located and matched across several admissions—a task where literal term overlap and semantic similarity are each informative but neither alone is sufficient. This complementarity is exactly what Hybrid exploits: fusing BM25 and EmbeddingGemma gives the best result (Hit@1 63.0% for single document, 71.7% for multi-document, ahead of either retriever alone in each), because a chunk one family’s ranking misses is often recovered by the other’s, letting the fused ranking retrieve the correct source more reliably than either signal on its own.

Cost of on-device deployment. The generator reaches ~4.4–4.6 tokens/s on the test device, confirming interactive latency without a server component. Table 10 provides the storage side of this cost for the 4,022-chunk, 100-patient corpus: index size ranges 12.4–16.2 MB and indexing time 0.64–3.50 s across the four stores compared.

What does the document layer add? ANAMNESIS evaluates RAG from the document level rather than from pre-extracted database text. During benchmark construction, each MIMIC-III admission is materialized as a separate PDF containing a structured admission header and the corresponding discharge-summary narrative. These PDFs are the actual inputs to the on-device RAG pipeline. At ingestion time, the application extracts their text locally, normalizes it, splits it into overlapping chunks, associates each chunk with its source-document metadata, and indexes the resulting representations for retrieval. At query time, retrieval is therefore performed over information obtained through the document-ingestion pipeline rather than directly over the original MIMIC-III tables. This makes document ingestion, chunking, indexing, and source attribution explicit components of the evaluated system while keeping the input interface independent of the database schema from which the benchmark was constructed.

7 Conclusion

We introduced ANAMNESIS, the first benchmark for patient-level RAG over longitudinal clinical records in document form, pairing open-ended questions with reference answers and chunk-level evidence ground truth—including cross-document longitudinal questions that make retrieval necessary by construction. Together with

a fully on-device reference system, ANAMNESIS enables the community to measure, diagnose, and improve personalized clinical information access under the privacy constraints that motivate it. Our experiments show that effective evidence selection is critical to on-device patient-level RAG: retrieved top- k context substantially outperforms full-folder prompting, while complementary lexical and semantic retrieval improves the evidence supplied to the generator. Yet, the remaining errors under oracle evidence show that the resource constraints of fully on-device execution expose a coupled retrieval–generation challenge, in which both evidence selection and the generator’s ability to use the available clinical context shape end-to-end performance. Future work includes cross-patient distractor settings, multilingual records, human-validated question realism, and extending the construction pipeline to MIMIC-IV-Note.

Limitations. ANAMNESIS inherits the demographic and institutional biases of MIMIC-III (a single U.S. critical-care cohort, 2001–2012) and the de-identification placeholders of its notes; in particular, the critical-care population differs from the outpatient mental-health scenario motivating our reference application, so absolute scores should not be read as estimates for that deployment. LLM-generated questions and reference answers, despite the automated answerability, grounding, and overlap checks of §3.3, have not yet been verified by clinicians; moreover, generation currently relies on a proprietary model, which is in tension with exact regeneration. Automatic answer evaluation relies on a single LLM judge (GPT-5.6 Sol), which may introduce evaluator-specific bias; we therefore complement judge-based correctness with an independent embedding-similarity measure (see §4). The benchmark measures information access over records, not clinical advice quality or safety. Finally, derived data remain governed by the PhysioNet Credentialed Health Data Use Agreement: we therefore release identifiers, code, and prompts enabling exact regeneration, rather than the raw text itself.

Ethical considerations. All source data are de-identified and accessed under the MIMIC-III DUA; no re-identification was attempted, and generated content was never crossed with external databases. The reference system is a research prototype: it must not be construed as a medical device, and answers grounded in personal records do not substitute professional consultation. On-device execution is itself a privacy measure—no protected content leaves the user’s device—but rigorous clinical validation would be required before any patient-facing deployment.

GenAI Usage Disclosure

Generative AI (GenAI) tools were utilized in the preparation of this manuscript to assist with coding and language polishing. The core research ideas, experimental design, analysis, and the overall drafting and writing of the paper were conducted by the authors without the involvement of GenAI.

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